

1. (Question 1). Let  $X$  be a discrete random variable with the given probability mass function with  $0 \leq \theta \leq 1$  as a parameter:

| $x$                 | 0           | 1          | 2                 | 3                |
|---------------------|-------------|------------|-------------------|------------------|
| $P(X = x   \theta)$ | $2\theta/3$ | $\theta/3$ | $2(1 - \theta)/3$ | $(1 - \theta)/3$ |

We have the following i.i.d. sample  $\mathbf{x} = (3, 0, 2, 1, 3, 2, 1, 0, 2, 1)$  from  $X$  and we want the maximum likelihood estimate of  $\theta$ . Let  $x^t$  denote the  $t$ -th observation, for  $t = 1, \dots, N$ , where here  $N = 10$ , so that  $\mathbf{x} = (x^1, \dots, x^N)$ . Since the observations are independent, the likelihood of the parameter  $\theta$  is

$$\begin{aligned} l(\theta | \mathbf{x}) &= \prod_{t=1}^N p(x^t | \theta) \\ &= p(3 | \theta) \cdot p(0 | \theta) \cdot p(2 | \theta) \cdot p(1 | \theta) \cdot p(3 | \theta) \\ &\quad \cdot p(2 | \theta) \cdot p(1 | \theta) \cdot p(0 | \theta) \cdot p(2 | \theta) \cdot p(1 | \theta) \\ &= \frac{1 - \theta}{3} \cdot \frac{2\theta}{3} \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \cdot \frac{1 - \theta}{3} \\ &\quad \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \cdot \frac{2\theta}{3} \cdot \frac{2(1 - \theta)}{3} \cdot \frac{\theta}{3} \\ &= \left(\frac{2\theta}{3}\right)^2 \left(\frac{\theta}{3}\right)^3 \left(\frac{2(1 - \theta)}{3}\right)^3 \left(\frac{1 - \theta}{3}\right)^2 \\ &= \frac{2^5}{3^{10}} \theta^5 (1 - \theta)^5. \end{aligned}$$

We can optimize via the log likelihood and our desired MLE for  $\theta$  is  $\hat{\theta} = \arg \max_{\theta} (L(\theta | \mathbf{x}))$  so consider

$$\begin{aligned} L(\theta | \mathbf{x}) &= \log(l(\theta | \mathbf{x})) \\ &= \log\left(\frac{2^5}{3^{10}} \theta^5 (1 - \theta)^5\right). \end{aligned}$$

Applying rules of logarithms we have

$$L(\theta | \mathbf{x}) = \log\left(\frac{2^5}{3^{10}}\right) + 5 \log \theta + 5 \log(1 - \theta).$$

Taking the derivative of  $L$  with respect to  $\theta$

$$\begin{aligned}\frac{dL}{d\theta} &= 0 + 5 \cdot \frac{1}{\theta} + 5 \cdot \frac{1}{1-\theta} \cdot (-1) \\ &= \frac{5}{\theta} - \frac{5}{1-\theta}.\end{aligned}$$

Set the derivative equal to zero and solve for  $\theta$ :

$$\begin{aligned}0 &= \frac{5}{\theta} - \frac{5}{1-\theta} \\ \frac{5}{1-\theta} &= \frac{5}{\theta} \\ \theta &= 1-\theta \\ \theta &= \frac{1}{2}.\end{aligned}$$

To verify that this gives a maximum, rewrite the derivative with a common denominator:

$$\begin{aligned}\frac{dL}{d\theta} &= \frac{5}{\theta} - \frac{5}{1-\theta} \\ &= \frac{5(1-\theta) - 5\theta}{\theta(1-\theta)} \\ &= \frac{5(1-2\theta)}{\theta(1-\theta)}.\end{aligned}$$

For  $0 < \theta < 1$ , the denominator is positive, so the sign of the derivative is determined by  $1 - 2\theta$ . Thus the derivative is positive for  $0 < \theta < \frac{1}{2}$ , zero at  $\theta = \frac{1}{2}$ , and negative for  $\frac{1}{2} < \theta < 1$ . The log likelihood therefore increases up to  $\theta = \frac{1}{2}$  and decreases afterward. Since the logarithm is strictly increasing, this also maximizes the likelihood on  $(0, 1)$ . At the endpoints  $\theta = 0$  and  $\theta = 1$ , the likelihood is zero, so neither endpoint gives a larger value.

Therefore the maximum likelihood estimate of  $\theta$  is  $\hat{\theta} = \boxed{\frac{1}{2}}$ .

2. (*Question 2*). Let  $X_1, X_2, \dots, X_N$  be i.i.d. random variables with density function

$$f(x \mid \sigma) = \frac{1}{2\sigma} e^{-\frac{|x|}{\sigma}}.$$

Again, since the observations are independent, we define the likelihood function as

$$\begin{aligned}
l(\sigma \mid x) &\equiv p(x \mid \sigma) = \prod_{t=1}^N p(x_t \mid \sigma) \\
&= \prod_{t=1}^N f(x_t \mid \sigma) \\
&= \prod_{t=1}^N \left( \frac{1}{2\sigma} e^{-\frac{|x_t|}{\sigma}} \right) \\
&= \left( \frac{1}{2\sigma} \right)^N \prod_{t=1}^N \left( e^{-\frac{|x_t|}{\sigma}} \right) \\
&= \left( \frac{1}{2\sigma} \right)^N e^{\left( \sum_{t=1}^N -\frac{|x_t|}{\sigma} \right)} \\
&= \left( \frac{1}{2\sigma} \right)^N e^{\left( -\frac{1}{\sigma} \sum_{t=1}^N |x_t| \right)}.
\end{aligned}$$

We then take the log, which we will optimize.

$$\begin{aligned}
L(\sigma \mid x) &= \log(l(\sigma \mid x)) \\
&= \log \left[ \left( \frac{1}{2\sigma} \right)^N e^{-\frac{1}{\sigma} \sum_{t=1}^N |x_t|} \right].
\end{aligned}$$

Applying rules of logarithms we have

$$\begin{aligned}
L(\sigma \mid x) &= \log \left[ \left( \frac{1}{2\sigma} \right)^N \right] + \log \left[ e^{-\frac{1}{\sigma} \sum_{t=1}^N |x_t|} \right] \\
&= N \log \left( \frac{1}{2\sigma} \right) - \frac{1}{\sigma} \sum_{t=1}^N |x_t| \\
&= -N \log(2\sigma) - \frac{1}{\sigma} \sum_{t=1}^N |x_t| \\
&= -N \log 2 - N \log \sigma - \frac{1}{\sigma} \sum_{t=1}^N |x_t|.
\end{aligned}$$

Taking the derivative of  $L$  with respect to  $\sigma$

$$\begin{aligned}
\frac{dL}{d\sigma} &= 0 - N \cdot \frac{1}{\sigma} - \left(-\frac{1}{\sigma^2}\right) \sum_{t=1}^N |x_t| \\
&= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t|.
\end{aligned}$$

Set the derivative equal to zero and solve for  $\sigma$ :

$$\begin{aligned}
0 &= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t| \\
\frac{N}{\sigma} &= \frac{1}{\sigma^2} \sum_{t=1}^N |x_t|
\end{aligned}$$

Since  $\sigma > 0$ , we have  $\sigma^2 > 0$ , so we can multiply both sides by  $\sigma^2$ :

$$\begin{aligned}
N\sigma &= \sum_{t=1}^N |x_t| \\
\sigma &= \frac{1}{N} \sum_{t=1}^N |x_t|.
\end{aligned}$$

To verify that this gives a maximum, let  $\hat{\sigma} = \frac{1}{N} \sum_{t=1}^N |x_t|$  and rewrite the derivative with a common denominator:

$$\begin{aligned}
\frac{dL}{d\sigma} &= -\frac{N}{\sigma} + \frac{1}{\sigma^2} \sum_{t=1}^N |x_t| \\
&= \frac{\sum_{t=1}^N |x_t| - N\sigma}{\sigma^2} \\
&= \frac{N\hat{\sigma} - N\sigma}{\sigma^2} \\
&= \frac{N(\hat{\sigma} - \sigma)}{\sigma^2}.
\end{aligned}$$

For  $\sigma > 0$ , the factor  $N/\sigma^2$  is positive, so the sign of the derivative is determined by  $\hat{\sigma} - \sigma$ . When  $\sum_{t=1}^N |x_t| > 0$ , we have  $\hat{\sigma} > 0$ . Thus the derivative is positive for  $0 < \sigma < \hat{\sigma}$ , zero at  $\sigma = \hat{\sigma}$ , and negative for  $\sigma > \hat{\sigma}$ . The log likelihood therefore increases up to  $\hat{\sigma}$  and decreases afterward, so this gives the maximum. Since the logarithm is strictly increasing, maximizing the likelihood is equivalent to maximizing its log. Therefore the maximum likelihood estimate of  $\sigma$  is

$$\begin{aligned}
\hat{\sigma} &= \arg \max_{\sigma > 0} l(x \mid \sigma) \\
&= \arg \max_{\sigma > 0} \log (l(x \mid \sigma)) \\
&= \arg \max_{\sigma > 0} L(\sigma \mid x) \\
&= \boxed{\frac{1}{N} \sum_{t=1}^N |x_t|}.
\end{aligned}$$

3. (*Question 3*). We are given  $N$  random samples  $X_1, X_2, \dots, X_N$  where we assume that the density function is the Gaussian:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

We use MLE to estimate the parameters  $\mu$  and  $\sigma$ . I prefer  $e$  notation, but the exponents are getting very large for this problem so I am going to switch to  $\exp$ :

$$\begin{aligned}
l(\mu, \sigma \mid x) &\equiv p(x \mid \mu, \sigma) = \prod_{t=1}^N p(x_t \mid \mu, \sigma) \\
&= \prod_{t=1}^N f(x_t \mid \mu, \sigma^2) \\
&= \prod_{t=1}^N \left[ \frac{1}{\sqrt{2\pi}\sigma} \exp \left( -\frac{(x_t - \mu)^2}{2\sigma^2} \right) \right] \\
&= \left( \frac{1}{\sqrt{2\pi}\sigma} \right)^N \prod_{t=1}^N \exp \left( -\frac{(x_t - \mu)^2}{2\sigma^2} \right) \\
&= \left( \frac{1}{\sqrt{2\pi}\sigma} \right)^N \exp \left[ \sum_{t=1}^N \left( -\frac{(x_t - \mu)^2}{2\sigma^2} \right) \right] \\
&= \frac{1}{(2\pi)^{N/2} \sigma^N} \exp \left[ -\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \right].
\end{aligned}$$

This is old hat at this point, we take the log to make differentiation easier.

$$\begin{aligned}
L(\mu, \sigma \mid x) &= \log(l(\mu, \sigma \mid x)) \\
&= \log \left[ \frac{1}{(2\pi)^{N/2} \sigma^N} \exp \left( -\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \right) \right] \\
&= \log \left( \frac{1}{(2\pi)^{N/2} \sigma^N} \right) + \log \left[ \exp \left( -\frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \right) \right] \\
&= -\log((2\pi)^{N/2} \sigma^N) - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\log((2\pi)^{N/2}) - \log(\sigma^N) - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\frac{N}{2} \log(2\pi) - N \log \sigma - \frac{1}{2\sigma^2} \sum_{t=1}^N (x_t - \mu)^2.
\end{aligned}$$

Since we are estimating for both  $\mu$  and  $\sigma$  and  $L$  is a function of both, we will have to take partial derivatives. First with respect to  $\mu$ :

$$\begin{aligned}
\frac{\partial L}{\partial \mu} &= 0 - 0 - \frac{1}{2\sigma^2} \sum_{t=1}^N 2(x_t - \mu) \cdot (-1) \\
&= \frac{1}{\sigma^2} \sum_{t=1}^N (x_t - \mu) \\
&= \frac{1}{\sigma^2} \left( \sum_{t=1}^N x_t - N\mu \right).
\end{aligned}$$

Then with respect to  $\sigma$ :

$$\begin{aligned}
\frac{\partial L}{\partial \sigma} &= 0 - N \cdot \frac{1}{\sigma} - \frac{1}{2} \cdot (-2\sigma^{-3}) \sum_{t=1}^N (x_t - \mu)^2 \\
&= -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \mu)^2.
\end{aligned}$$

Now we set these partial derivatives equal to 0 and solve.

$$0 = \frac{1}{\sigma^2} \left( \sum_{t=1}^N x_t - N\mu \right)$$

Since  $\sigma > 0$ , we can multiply both sides by  $\sigma^2$ :

$$\begin{aligned} 0 &= \sum_{t=1}^N x_t - N\mu \\ N\mu &= \sum_{t=1}^N x_t \\ \mu &= \boxed{\frac{1}{N} \sum_{t=1}^N x_t = \bar{x}}. \end{aligned}$$

For  $\sigma$ , substituting  $\mu = \bar{x}$  gives

$$\begin{aligned} 0 &= -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \bar{x})^2 \\ \frac{N}{\sigma} &= \frac{1}{\sigma^3} \sum_{t=1}^N (x_t - \bar{x})^2 \end{aligned}$$

Since  $\sigma > 0$ , we can multiply both sides by  $\sigma^3$ :

$$\begin{aligned} N\sigma^2 &= \sum_{t=1}^N (x_t - \bar{x})^2 \\ \sigma^2 &= \frac{1}{N} \sum_{t=1}^N (x_t - \bar{x})^2 \end{aligned}$$

Taking the positive square root, provided the samples are not all identical:

$$\sigma = \boxed{\sqrt{\frac{1}{N} \sum_{t=1}^N (x_t - \bar{x})^2}}.$$

These are, of course, the familiar formulas when working with the normal distribution from statistics.

4. (*Question 4*). Let  $X_1, X_2, \dots, X_N$  be an i.i.d. sample from a Poisson distribution with parameter  $\lambda$  and probability mass function

$$P(X = x \mid \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

We use MLE to estimate  $\lambda$ . Again, since the observations are independent, the likelihood is

$$\begin{aligned}
l(\lambda \mid x) &= \prod_{t=1}^N p(x_t \mid \lambda) \\
&= \prod_{t=1}^N \frac{\lambda^{x_t} e^{-\lambda}}{x_t!} \\
&= \frac{\left( \prod_{t=1}^N \lambda^{x_t} \right) \left( \prod_{t=1}^N e^{-\lambda} \right)}{\prod_{t=1}^N x_t!} \\
&= \frac{\lambda^{\sum_{t=1}^N x_t} e^{-N\lambda}}{\prod_{t=1}^N x_t!}.
\end{aligned}$$

We then take the log to make differentiation easier. For  $\lambda > 0$ , applying rules of logarithms gives

$$\begin{aligned}
L(\lambda \mid x) &= \log(l(\lambda \mid x)) \\
&= \log\left(\frac{\lambda^{\sum_{t=1}^N x_t} e^{-N\lambda}}{\prod_{t=1}^N x_t!}\right) \\
&= \log\left(\lambda^{\sum_{t=1}^N x_t}\right) + \log(e^{-N\lambda}) - \log\left(\prod_{t=1}^N x_t!\right) \\
&= \left(\sum_{t=1}^N x_t\right) \log \lambda - N\lambda - \sum_{t=1}^N \log(x_t!).
\end{aligned}$$

Taking the derivative of  $L$  with respect to  $\lambda$

$$\begin{aligned}
\frac{dL}{d\lambda} &= \left(\sum_{t=1}^N x_t\right) \cdot \frac{1}{\lambda} - N - 0 \\
&= \frac{1}{\lambda} \sum_{t=1}^N x_t - N.
\end{aligned}$$

Set the derivative equal to zero and solve for  $\lambda$ :

$$\begin{aligned}
0 &= \frac{1}{\lambda} \sum_{t=1}^N x_t - N \\
N &= \frac{1}{\lambda} \sum_{t=1}^N x_t
\end{aligned}$$

Since  $\lambda > 0$ , we can multiply both sides by  $\lambda$ :

$$N\lambda = \sum_{t=1}^N x_t$$

$$\lambda = \frac{1}{N} \sum_{t=1}^N x_t = \bar{x}.$$

To verify that this gives a maximum when  $\bar{x} > 0$ , rewrite the derivative as

$$\frac{dL}{d\lambda} = \frac{\sum_{t=1}^N x_t - N\lambda}{\lambda}$$

$$= \frac{N(\bar{x} - \lambda)}{\lambda}.$$

For  $\lambda > 0$ , the factor  $N/\lambda$  is positive, so the sign of the derivative is determined by  $\bar{x} - \lambda$ . Thus the derivative is positive for  $0 < \lambda < \bar{x}$ , zero at  $\lambda = \bar{x}$ , and negative for  $\lambda > \bar{x}$ . The log likelihood therefore increases up to  $\bar{x}$  and decreases afterward. Since the logarithm is strictly increasing, this also maximizes the likelihood.

Therefore, on the parameter space  $\lambda > 0$ , the maximum likelihood estimate is

$$\hat{\lambda} = \arg \max_{\lambda > 0} l(\lambda \mid x)$$

$$= \arg \max_{\lambda > 0} L(\lambda \mid x)$$

$$= \boxed{\frac{1}{N} \sum_{t=1}^N x_t = \bar{x}}.$$

5. (*Question 5*). We are given  $N$  data samples  $(X_1, Y_1), (X_2, Y_2), \dots, (X_N, Y_N)$ . For each input  $X_i$ , the neural network predicts both a mean  $\mu(X_i)$  and a standard deviation  $\sigma(X_i)$ . We assume that the output  $Y$  follows a Gaussian distribution conditioned on  $X$ :

$$Y \mid X \sim \mathcal{N}(\mu(X), \sigma^2(X)).$$

Therefore, for a single sample  $(X_i, Y_i)$ , the conditional density is

$$p(Y_i \mid X_i) = \frac{1}{\sqrt{2\pi}\sigma(X_i)} \exp\left(-\frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)}\right).$$

As before, assuming the samples are independent, the likelihood of observing all  $N$  samples is the product of the individual probabilities:

$$\begin{aligned}
l &= \prod_{i=1}^N p(Y_i | X_i) \\
&= \prod_{i=1}^N \left[ \frac{1}{\sqrt{2\pi}\sigma(X_i)} \exp \left( -\frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right) \right].
\end{aligned}$$

We could maximize this likelihood directly, but as before it is easier to work with the log likelihood:

$$\begin{aligned}
L &= \log l \\
&= \sum_{i=1}^N \log \left[ \frac{1}{\sqrt{2\pi}\sigma(X_i)} \exp \left( -\frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right) \right] \\
&= \sum_{i=1}^N \left[ \log \left( \frac{1}{\sqrt{2\pi}\sigma(X_i)} \right) - \frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right] \\
&= \sum_{i=1}^N \left[ -\frac{1}{2} \log(2\pi) - \log \sigma(X_i) - \frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right].
\end{aligned}$$

Maximum Likelihood Estimation seeks to maximize this quantity. However, the loss function for the neural network is to be minimized, so we instead minimize the negative log likelihood:

$$\mathcal{L} = -L.$$

Therefore,

$$\mathcal{L} = \sum_{i=1}^N \left[ \frac{1}{2} \log(2\pi) + \log \sigma(X_i) + \frac{(Y_i - \mu(X_i))^2}{2\sigma^2(X_i)} \right].$$

Equivalently, using  $\log(\sigma^2) = 2 \log \sigma$ ,

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^N \left[ \log(2\pi\sigma^2(X_i)) + \frac{(Y_i - \mu(X_i))^2}{\sigma^2(X_i)} \right].$$

6. (Question 6). Stacked deep decision trees for activity classification.

**Implementation:** See `q6_cascade_forest.ipynb` for the AI-assisted code, experiment outputs, and learning prompts. AI coding assistance is permitted by this question.

**Architecture.** *[Explain the four forests in each layer, their class-probability outputs, concatenation with the original features, and the final averaging and class selection. Note how the randomized forests are implemented.]*

**Experimental setup.** *[Describe the UCI HAR data, the 80:10:10 split, out-of-fold training features, and validation-based depth selection. Explain what each split is used for.]*

**Results.** *[Use the notebook outputs to report the selected depth, both test accuracies and macro F1 scores, and the accuracy difference in percentage points. Discuss the confusion matrices and training cost.]*

**Discussion.** *[Explain whether stacking helped in this run. Discuss participant overlap across splits, the single split and seed, and the difference in computational cost. State what the experiment does and does not establish.]*